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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

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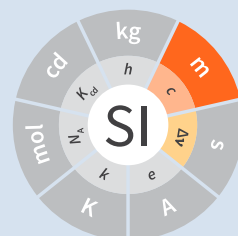
Ytterbium 171 Ion ()

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CCL-CCTF Frequency Standards Working Group

April 13, 2022

13 avril 2022



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Ytterbium 171 Ion ()**

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1. Recommended value [CCTF 22nd Meeting \(2020\)](#) of the frequency in the CIPM List of Frequencies

$$f(^{171}\text{Yb}^+, \text{quadrupole}) = 688\,358\,979\,309\,308.24 \text{ Hz}$$

equivalent to

$$\lambda(^{171}\text{Yb}^+) = 435\,517\,610.739\,687\,63 \text{ fm}$$

with a relative standard uncertainty of 2.0×10^{-16} .

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second [CIPM REC 1 \(2006, E\)](#).

2. Method to establish the recommended value

A global adjustment of all measurements of frequency ratios published in peer-reviewed publications and available to the CCL-CCTF WGFS was carried out following the methods presented in [3] to [7].

This adjustment determines the frequency of 14 transitions (see Figure 1, see p. 5) which are either already adopted as secondary representations of the second [7] or considered as candidates for SRS. It took into account 105 measurements, including 33 frequency ratios and 72 absolute frequency measurements (i.e. ratios to the ^{133}Cs frequency). A total of 483 correlations between these input measurements were estimated and considered in the adjustment. More details on the input data and the processing are provided at https://webtai.bipm.org/ftp/pub/tai/publication/wgfs/Adjustment_2021.html. The recommended value is the direct result of the adjustment, rounded as deemed adequate with respect to the recommended uncertainty.

While the results are from a global adjustment, it is of interest to note (see Figure 1, see p. 5) that the $^{171}\text{Yb}^+$ transition is involved in 5 measurements relative to ^{133}Cs , and in 2 frequency ratios with an optical transition.

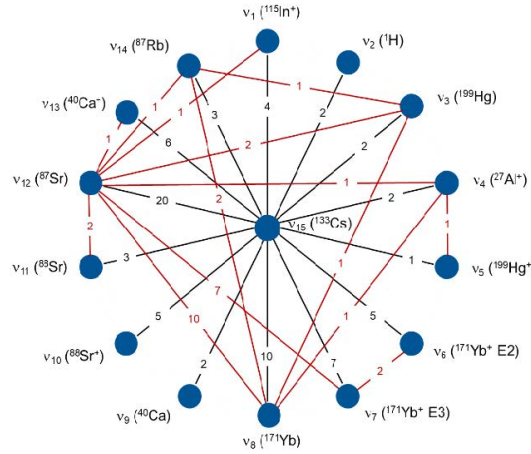


Figure 1 — Representation of the 105 measurements linking 14 transitions on the circle and ^{133}Cs at the center.

$^{171}\text{Yb}^+$ ion, $6s^2S_{1/2}$ ($F=0, mF=0$)– $5d^2D_{3/2}$ ($F=2, mF=0$) unperturbed optical transition

References

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- [2] CIPM REC 1 (2006, E) International Committee for Weights and Measures (2006) *Concerning secondary representations of the second* (Paris: International Bureau of Weights and Measures) <https://www.bipm.org/en/committees/ci/cipm/95-2006/resolution-1>.
- [3] H S Margolis and P Gill (2015) Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values. *Metrologia*. **52** (5) 628–634. <https://doi.org/10.1088/0026-1394/52/5/628>.
- [4] L Robertsson (2016) On the evaluation of ultra-high-precision frequency ratio measurements: examining closed loops in a graph theory framework. *Metrologia*. **53** (6) 1272–1280. <https://doi.org/10.1088/0026-1394/53/6/1272>.
- [5] G. Panfilo, communication to the CCL-CCTF WGFS. A new implementation of [4] was realized in MatLab at the BIPM (2020)
- [6] Ch. Oates, communication to the CCL-CCTF WGFS. An independent program was developed in Mathematica at NIST (2017)
- [7] Fritz Riehle, Patrick Gill, Felicitas Arias and Lennart Robertsson (2018) The CIPM list of recommended frequency standard values: guidelines and procedures. *Metrologia*. **55** (2) 188–200. <https://doi.org/10.1088/1681-7575/aaa302>.